

TABLE 1
Summary of expected properties for each simulation design

	Parallel Trends		Pulse Pre-Trend	
	$\Delta^{\text{SD}}(M)$	$\Delta^{\text{SDPB}}(M)$	$\Delta^{\text{SDRM}}(\bar{M})$	$\Delta^{\text{RM}}(\bar{M})$
Conditional and Hybrid				
Consistent	✓	✓	✓	✓
Asymptotically (near-)optimal	✓	✓	✓	×
FLCI				
Consistent	✓	×	×	×
Finite-sample near-optimal	✓	×	×	×

is the average causal effect in the post-treatment periods ($\theta = \bar{\tau}_{\text{post}}$), with qualitatively similar results.

For a given choice of Δ , we compute the identified set $\mathcal{S}(\beta, \Delta)$ using the formulas provided in Lemma 2.1 and calculate the expected excess length for each of the proposed confidence sets. We benchmark the expected excess length of our proposed confidence sets relative to an efficiency bound for confidence sets that satisfy the uniform coverage requirement.²⁹ We report the efficiency ratio of each procedure, which is defined as the ratio of the optimal benchmark relative to the average excess length for the procedure. All results are calculated over 1,000 simulations per paper.

We consider four choices of Δ to highlight the performance of our proposed confidence sets across a range of conditions: $\Delta^{\text{SD}}(M)$, $\Delta^{\text{SDPB}}(M)$, $\Delta^{\text{RM}}(\bar{M})$, and $\Delta^{\text{SDRM}}(\bar{M})$. We consider simulations under the assumption of zero treatment effects, so that $\tau = 0$ and thus $\beta = \delta$. We consider two forms for δ . First, we consider the baseline case of parallel trends ($\delta = 0$). Second, we consider a “pulse” pre-trend in which δ_{-1} is non-zero and the remaining elements of δ are zero. Such a pre-trend might arise in practice if there are confounding policy changes or other events close to the time of treatment. These different choices of δ allow us to highlight the relative strengths of the proposed inference procedures. For example, FLCIs have near-optimal expected length when $\delta = 0$ and $\Delta = \Delta^{\text{SD}}(M)$, whereas the conditional test has optimal local asymptotic power under the pulse design when $\Delta = \Delta^{\text{SDPB}}(M)$. Table 1 summarizes which of our theoretical results hold for each of the simulation designs when M and $\bar{M}$ are non-zero.

In practice, we find that for $\Delta^{\text{SD}}(M)$ and $\Delta^{\text{SDPB}}(M)$, the results depend on M but are qualitatively similar across values of δ . By contrast, for $\Delta^{\text{SDRM}}(\bar{M})$ and $\Delta^{\text{RM}}(\bar{M})$, the choice of δ is more important than the choice of $\bar{M}$. Therefore, to highlight the most important dimensions for each of the simulation designs, in the main text of the paper we report results for $\Delta^{\text{SD}}(M)$ and $\Delta^{\text{SDPB}}(M)$ under different values of M and $\delta = 0$ (parallel trends), whereas for $\Delta^{\text{RM}}(\bar{M})$ and $\Delta^{\text{SDRM}}(\bar{M})$ we vary the magnitude of the pre-treatment pulse δ_{-1} , holding $\bar{M} = 1$ constant. In the [supplementary materials](#), we report results for additional choices of these parameters.

29. For choices of Δ that are convex (e.g. $\Delta^{\text{SD}}(M)$ and $\Delta^{\text{SDPB}}(M)$), we benchmark the expected excess length of our proposed confidence sets against a sharp optimal bound over confidence sets that satisfy the finite-sample coverage requirement (10). This optimal bound is provided in the [supplementary materials](#), and follows as a corollary from results in [Armstrong and Kolesár \(2018\)](#) on the optimal expected length of a confidence set satisfying the uniform coverage requirement (10). For choices of Δ that can be written as the union of convex sets (e.g. $\Delta^{\text{RM}}(\bar{M})$ and $\Delta^{\text{SDRM}}(\bar{M})$), we compare the expected excess length of our proposed confidence sets against the maximal optimal bound over each set in the union, which is a potentially non-sharp bound for any confidence set with correct coverage.

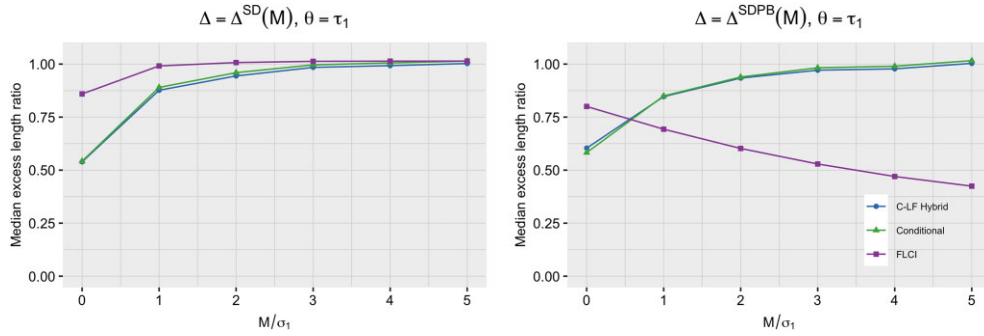


FIGURE 2

Simulation results for $\Delta^{\text{SD}}(M)$ and $\Delta^{\text{SDPB}}(M)$: Median efficiency ratios for proposed procedures.

Note: Median efficiency ratios for our proposed confidence sets over $\Delta^{\text{SD}}(M)$ and $\Delta^{\text{SDPB}}(M)$ under the assumption of parallel trends and zero treatment effects (*i.e.* $\beta = 0$). The efficiency ratio for a procedure is defined as the efficiency bound divided by the procedure's expected excess length. The results for the FLCI are plotted in purple, conditional-LF ("C-LF Hybrid") hybrid in blue, and conditional confidence set in green. Results are averaged over 1,000 simulations for each of the 12 papers surveyed, and the median across papers is reported here.

We report results for three methods for constructing confidence sets: FLCIs, conditional confidence sets, and conditional-least favourable hybrid confidence sets.³⁰ For $\Delta^{\text{RM}}(\bar{M})$ and $\Delta^{\text{SDRM}}(\bar{M})$, we omit results for the FLCI since the FLCIs have infinite length.

5.2. Simulation results

To compare results easily across the 12 papers in the simulation study, we normalize the units of δ_{-1} and M by the standard deviation of $\hat{\beta}_1$ (denoted σ_1). Large normalized values of M or δ_{-1} correspond with the case where the identified set is large relative to sampling variation, mimicking our asymptotic power results in which sampling variation grows small relative to the identified set. In the graphs below, we report the median value of excess length efficiency across the papers in the survey. The normalization described above implies that the units of the x -axis correspond with the worst-case bias of the naive estimator $\hat{\beta}_1$ divided by its standard error.³¹

5.2.1. Results for $\Delta^{\text{SD}}(M)$. The left panel of Figure 2 plots the efficiency ratio for each procedure as a function of M/σ_1 when $\Delta = \Delta^{\text{SD}}(M)$. All procedures perform well as M/σ_1 grows large with efficiency ratios approaching 1, illustrating our asymptotic (near-)optimality results for this design. However, the FLCIs perform best for smaller values of M/σ_1 , including the point-identified case where $M = 0$, illustrating the finite-sample near-optimality results for the FLCIs when Assumption 8 holds. Although the conditional and hybrid confidence sets have efficiency approaching the optimal bound for M/σ_1 large, their efficiency is only about 50% when $M/\sigma_1 = 0$, in which case θ is point identified and thus LICQ does not hold. The conditional and hybrid confidence sets perform similarly.

30. For the conditional-least favourable hybrid confidence sets, we use a first-stage least-favourable test of size $\kappa = \alpha/10$, following ARP and Romano et al. (2014).

31. For $\hat{\beta}_1$ normally distributed, the worst-case coverage of a conventional 95% confidence interval as a function of the normalized worst-case bias b is $\Phi(1.96 + b) - \Phi(-1.96 + b)$, which is 0.95 for $b = 0$, 0.83 for $b = 1$, 0.48 for $b = 2$, etc.

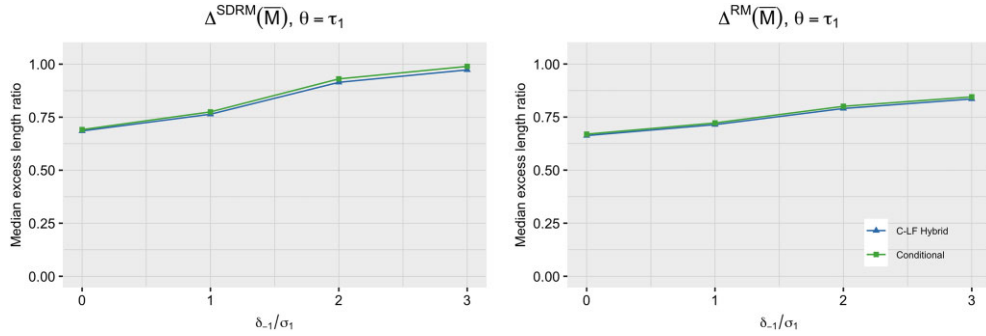


FIGURE 3

Simulation results for $\Delta^{SDRM}(\bar{M})$ and $\Delta^{RM}(\bar{M})$: Median efficiency ratios for proposed procedures.

Note: Median efficiency ratios for our proposed confidence sets over $\Delta^{SDRM}(\bar{M})$ and $\Delta^{RM}(\bar{M})$ with $\bar{M} = 1$ under the assumption of zero treatment effects and a “pulse” pre-trend (*i.e.* $\beta_{-1} = \delta_{-1}$ and $\beta_t = 0$ for all $t \neq -1$). The efficiency ratio for a procedure is defined as the efficiency bound divided by the procedure’s expected excess length. The results for the conditional-least favourable (“C-LF”) hybrid are plotted in blue, and conditional confidence set in green. Results are averaged over 1,000 simulations for each of the 12 papers surveyed, and the median across papers is reported here.

5.2.2. Results for $\Delta^{SDPB}(\bar{M})$. The right panel of Figure 2 plots the efficiency ratio for each procedure as a function of M/σ_1 when $\Delta = \Delta^{SDPB}(\bar{M})$. The efficiency ratios for the conditional and hybrid confidence sets are again (near-)optimal as M/σ_1 grows large, highlighting our asymptotic (near-)optimality results for these procedures in this simulation design. By contrast, the efficiency ratios for the FLCIs steadily decrease as M/σ_1 increases, reflecting that the FLCIs are not consistent in this simulation design when $M > 0$. The conditional-LF hybrid confidence sets slightly improve efficiency relative to the conditional confidence sets when M/σ_1 is small and retain near-optimal performance as M/σ_1 grows large.

5.2.3. Results for $\Delta^{SDRM}(\bar{M})$. The left panel of Figure 3 plots the efficiency ratios for the conditional and conditional-least favourable hybrid confidence sets as a function of δ_{-1}/σ_1 when $\Delta = \Delta^{SDRM}(\bar{M})$. We omit results for the optimal FLCI since the optimal FLCI has infinite length for this design. Both procedures perform well as δ_{-1}/σ_1 grows large with efficiency ratios approaching 1, illustrating our asymptotic (near-) optimality result for this design. Both procedures also have similar power curves, with slightly higher power for the conditional.

5.2.4. Results for $\Delta^{RM}(\bar{M})$. The right panel of Figure 3 plots the efficiency ratio for the conditional and conditional-least favourable hybrid confidence sets as a function of δ_{-1}/σ_1 when $\Delta = \Delta^{RM}(\bar{M})$. We again omit results for the optimal FLCI since the optimal FLCI has infinite length for this design. The conditions for our asymptotic (near-) optimality result for unions of convex sets do not hold in this simulation design (as the maximum pre-period violation is not unique). Nonetheless, we find that the efficiency ratios for the conditional and hybrid confidence sets approach about 83% when δ_{-1}/σ_1 grow large. We find it somewhat encouraging that these procedures can have excess length within 17% of the optimum even in cases where LICQ fails. Once again, we also find that the conditional and conditional-least favourable hybrid have similar power.

5.2.5. Takeaways from simulations. Two clear patterns emerge from our simulations. First, the conditional and hybrid confidence sets perform well across a wide range of specifications, with particularly good power when the length of the identified set is large relative to sampling variation. Second, the FLCIs have the best performance for $\Delta^{SD}(\bar{M})$, particularly when

M is small, which aligns with the finite-sample near-optimality results in Section 4. However, FLCIs can perform quite poorly for other classes of Δ .

Overall, we therefore recommend to use the conditional-LF hybrid confidence sets for generic forms of Δ , and optimal FLCIs for the special case of $\Delta^{\text{SD}}(M)$ (or other special cases where the consistency/finite-sample near-optimality of FLCIs is guaranteed). Although the conditional and hybrid approaches perform similarly in our simulations, we recommend the hybrid approach in general based on the guidance provided in ARP. We implement these recommendations in our applications in the next section.

6. PRACTICAL GUIDANCE AND EMPIRICAL ILLUSTRATIONS

6.1. *Practical guidance*

We recommend that researchers use our methods to construct robust confidence intervals under restrictions on the possible violations of parallel trends Δ that are motivated by domain knowledge in their empirical setting. We also suggest that researchers report sensitivity analyses to illustrate the sensitivity of their causal conclusions to alternative assumptions on the possible violations of parallel trends.

6.1.1. Choice of Δ . The choice of Δ should be motivated by economic knowledge about the types of possible confounding factors that could produce non-parallel trends. We now provide some guidance on how the choice of Δ can be motivated by domain knowledge, highlighting some cases where our leading examples, $\Delta^{\text{RM}}(\bar{M})$ and $\Delta^{\text{SD}}(M)$, would be sensible choices.

In some empirical settings, researchers may be concerned about differential economic shocks to the treated and control groups that generate violations of parallel trends. If the researcher believes that the magnitude of these differential shocks in the post-treatment period is not too different from the magnitude in the pre-treatment period, then it may be reasonable to assume $\delta \in \Delta^{\text{RM}}(\bar{M})$, which explicitly bounds the relative magnitudes of violations of parallel trends in the post-treatment based on observed violations in the pre-treatment period. In other settings, researchers may be worried about violations of parallel trends that arise due to differences in smoothly evolving secular trends that differentially affect the treated and comparison groups. In this case, it may be reasonable to assume $\delta \in \Delta^{\text{SD}}(M)$, which explicitly bounds the extent to which the slope of the difference in trends can vary across consecutive periods. Economic knowledge may imply additional restrictions as well. For example, if the researcher knows of a confounding policy change that would have a positive effect on the outcome, then it is reasonable to further assume that post-treatment difference in trends must be positive (*i.e.* $\delta_t \geq 0$ for $t > 0$).

In our empirical applications below, we illustrate how domain knowledge about the types of possible violations of parallel trends can inform the choice of Δ . We encourage applied researchers to use such domain knowledge to inform the restrictions they impose on the possible choices of parallel trends in their context.

6.1.2. Choice of inference procedure. Based on our theoretical results and Monte Carlo simulations, we recommend the ARP hybrid confidence sets for generic, polyhedral forms of Δ . For the special case of $\Delta^{\text{SD}}(M)$ —or other choices of Δ for which the consistency and finite-sample-near-optimality of FLCIs are guaranteed—we recommend FLCIs. Our recommended choice of inference procedure is implemented in the HonestDiD R and Stata packages that